

Downstream processing for antimicrobial peptide production

Maria Priscila Almario^a, Carlos Manuel Ramírez^a, and Luis H. Reyes^b

^a*Grupo de Diseño de Productos y Procesos (GDPP), Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia,*

^b*Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia*

1 Introduction

Antimicrobial peptides (AMPs) have emerged as a focal point of research over recent years owing to their prospective utility as alternative therapeutic agents against multidrug-resistant pathogens. These inherently occurring, short-chain peptides exhibit a broad spectrum of antimicrobial activity against various microorganisms, including bacteria, fungi, viruses, and even cancer cells (Fox, 2013). The biotechnological synthesis of AMPs, leveraging recombinant DNA technology and microbial expression systems, has become a viable pathway for large-scale, economically efficient production of these crucial biomolecules (Li, 2011).

Downstream processing (DSP) constitutes a pivotal phase in the biotechnological fabrication of AMPs. This stage encompasses the retrieval and purification of the target peptide from the production milieu, often entailing a range of operations such as cell disruption, clarification, separation, and purification (Desai, 2008; Fig. 11.1). It is noteworthy that DSP can represent up to 80% of the total production expenditure for biopharmaceuticals, underscoring the necessity for optimization to ascertain the economic feasibility of AMP production (Jagschies et al., 2018).

This chapter aims to delineate the fundamental principles, methodologies, and hurdles inherent to DSP in the biotechnological synthesis of AMPs. The discourse will encompass the diverse techniques employed for cell disruption, clarification, and purification, along with the factors dictating the selection of these methodologies. Moreover, the distinctive challenges pertaining to the DSP of AMPs and the strategies deployed to surmount them will be explored. The chapter will also shed light on emerging technologies and prospective advancements in DSP for AMPs.

2 Cell disruption

Cell disruption is a crucial initial step in the DSP of AMPs, necessitating the release of intracellular constituents for subsequent purification. The selection of an appropriate cell disruption method is predicated on the cell type, scale of operation, and the sensitivity of the target peptides to mechanical and enzymatic forces. Techniques such as high-pressure homogenization (HPH), sonication, enzymatic lysis, and mechanical milling are employed, each with distinct advantages and limitations. The efficacy of cell disruption significantly impacts the yield and quality of the AMPs, hence a meticulous evaluation of the method employed is indispensable to ensuring the integrity and functionality of the extracted peptides (Desai, 2008; Rahman et al., 2022; Roca-Pinilla et al., 2022).

2.1 High-pressure homogenization

HPH stands as a robust method for cell disruption, pivotal in liberating intracellular AMPs (Middelberg, 2000; Yadav & Kale, 2020). The mechanism involves subjecting cell suspensions to elevated pressures, which, when abruptly released, yield shear forces potent enough to rupture cell membranes (Middelberg, 2000). The technique is revered for its scalability, making it conducive to both pilot and industrial-scale operations (Levy et al., 2021; Vinchhi et al., 2021).

The core of HPH lies in its ability to maintain the integrity and bioactivity of the released peptides, minimizing the risk of denaturation or degradation (Levy et al., 2021; Yadav & Kale, 2020). It's the interplay of pressure magnitude and exposure duration that dictates the efficacy of cell disruption, rendering a thorough optimization of operational parameters essential (Ding & Kan, 2017; Yadav & Kale, 2020). The selection of homogenization pressure, the number of passes

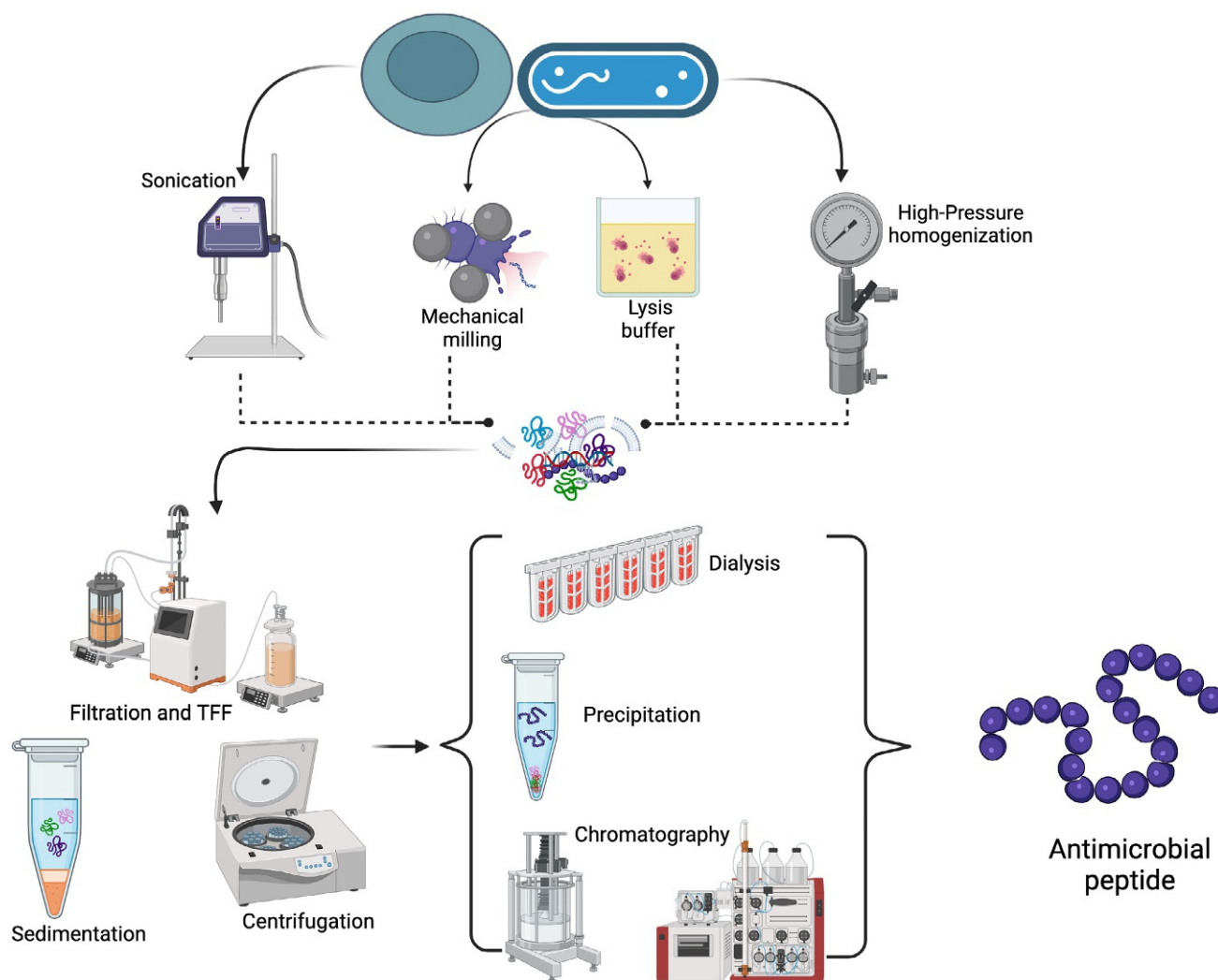


FIG. 11.1 Common steps in the downstream processing in the production of antimicrobial peptides. (Created with *BioRender.com*)

through the homogenizer, and controlling the temperature to mitigate heat generation are critical considerations to enhance the yield and purity of AMPs (Kruszewski et al., 2023; Levy et al., 2021).

HPH excels not only in cell disruption efficiency but also in rendering a homogeneous distribution of cell debris, which can be an advantageous prelude to downstream separation processes (Middelberg, 2000). Despite its merits, the energy demand (Xiao et al., 2023) and potential heat generation (Kruszewski et al., 2023; Levy et al., 2021) necessitate a meticulous evaluation to ensure the cost-effectiveness and suitability of HPH in the broader spectrum of AMP purification. Through adept optimization and integration within the DSP workflow, HPH can significantly contribute to the efficient and economical recovery of AMPs, aligning with the overarching goal of producing high-purity peptides for therapeutic applications (Dhiman et al., 2023).

2.2 Sonication

Sonication serves as a potent cell disruption technique, utilizing ultrasonic waves to generate mechanical vibrations and shock waves within a liquid medium (Saranya et al., 2014). These forces act upon the cell membranes, culminating in their rupture and subsequent release of intracellular contents, including AMPs. Key to sonication's efficacy is the optimization of parameters such as sonication duration, pulse intervals, and amplitude, which dictate the extent of cell lysis and the integrity of the liberated peptides (Ferdous et al., 2021; Pchelintsev et al., 2016; Saranya et al., 2014).

A notable advantage of sonication is its adaptability to varying scales, offering utility in both laboratory and industrial settings (Hielscher Ultrasonics GmbH, 2023a, 2023b, 2023c). Furthermore, sonication can be conducted in batch or continuous modes (Hielscher Ultrasonics GmbH, 2023b), enhancing its flexibility within the DSP framework. However, the technique's propensity to generate heat and potentially induce localized overheating warrants meticulous management of operational conditions to mitigate peptide denaturation risks (Jasmi et al., 2020; Saranya et al., 2014).

Sonication's capability to yield a homogenized suspension of cell debris provides a conducive milieu for subsequent separation and purification steps (Sun et al., 2018). The technique's integration within a comprehensive DSP strategy, when astutely managed, holds promise for augmenting the yield and purity of AMPs, thus aligning with the overarching objective of producing these bioactive molecules for therapeutic endeavors (Sun et al., 2018).

2.3 Enzymatic lysis

Enzymatic lysis, utilizing specific enzymes to dissolve the cell wall, stands as a gentler alternative to mechanical disruption methods (Rodríguez & Vanechoutte, 2019; Salazar & Asenjo, 2007). By targeting the structural components of the cell wall, enzymes such as lysozyme for bacteria or glucanase and chitinase for fungi, facilitate the release of intracellular contents, including AMPs (Rodríguez & Vanechoutte, 2019). The controlled nature of enzymatic lysis minimizes the risk of peptide degradation, albeit at a potentially slower rate compared to mechanical methods (Rodríguez & Vanechoutte, 2019).

The optimal conditions—enzyme concentration, incubation time, and temperature—need to be meticulously calibrated to ensure effective cell lysis while preserving the integrity of the AMPs. Further, the choice of enzymes is critical and is dictated by the cell wall composition of the microbial strain in use (Salazar & Asenjo, 2007).

The streamlined process of enzymatic lysis, characterized by fewer steps and fewer equipment requirements, offers an appealing approach, especially in scenarios where the AMPs are sensitive to mechanical or thermal stress (Adje et al., 2013). However, the enzyme cost and the necessity for precise condition optimization may pose challenges (Giovannoni et al., 2020). The integration of enzymatic lysis within the DSP regimen necessitates a careful balance between efficiency, cost, and the preservation of peptide activity, aligning with the broader objective of producing high-purity AMPs for therapeutic applications (Adje et al., 2013; Sun et al., 2018).

2.4 Mechanical milling

Mechanical milling employs abrasive forces to disintegrate the cellular structure, thereby liberating the intracellular contents, including AMPs (Maurin et al., 2017; Nemer et al., 2021). This method, using grinding media such as beads, exerts mechanical shear, and impact forces on the cells (Nemer et al., 2021). The process parameters—milling time, bead size, and rotational speed—require meticulous optimization to balance between efficient cell disruption and the preservation of peptide integrity (Kula & Schütte, 1987; Ramadhan & Foster, 2018).

Despite its high efficiency, mechanical milling may introduce contaminants from the grinding media, necessitating subsequent purification steps. Moreover, the generated heat and mechanical stress could potentially denature sensitive peptides (Harrison, 2011). Nonetheless, mechanical milling remains a viable option for large-scale cell disruption, given its scalability and ability to handle substantial biomass quantities (Goldberg, 2021). The selection of mechanical milling within a DSP framework should be judiciously aligned with the overarching goal of achieving high yields of bioactive AMPs in a cost-effective and time-efficient manner (Harrison, 2011).

3 Harvesting and recovery of antimicrobial peptides

In recent times, the advancement and refinement of purification methodologies have garnered increased attention, mainly due to the surge in bioproducts and bioprocesses, leading to an upswing in industrial bioprocesses (Jagschies et al., 2018). The pharmaceutical industry, along with other biotechnological sectors, grasps the significance of high-scale production capacity and its inherent limitations. As the production volume increases, the presence of impurities proportionally increases. The industry must adopt advanced harvesting and purification methodologies to ensure that raw materials align with regulatory specifications.

To navigate the challenges presented by increasing cell density and impurities in bioprocesses, cultures must be efficiently harvested and purified. The foundational harvesting methods have been established for decades and predominantly depend on mechanical solid-liquid separation techniques. Factors such as density, particle size, and electromagnetic

properties are frequently leveraged in these mechanical methods. Centrifugation, sedimentation, and filtration operations are particularly relevant to the pharmaceutical and biotechnological (Rios et al., 2007).

In this section, we will delve into various methods for harvesting and recovering AMPs from bioprocesses. We will discuss the principles, advantages, and limitations of mechanical solid-liquid separation techniques, including centrifugation, sedimentation, and filtration. Moreover, we will examine the factors influencing the choice of these methods and the challenges associated with their application in the context of AMP production.

3.1 Centrifugation

Centrifugation, as a methodology for retrieving extracellular products, hinges on the production platform utilized and the capability to expel the product beyond the cellular membrane. In industrial scenarios, centrifugation predominantly serves for the recovery of supernatants through in-line processes. Numerous considerations arise when orchestrating industrial in-line centrifuges, including the kind and magnitude of the centrifuge, rotor selection, and prime process parameters such as centrifugal force, duration, and temperature.

Companies such as Thermo Fisher Scientific, Beckman Coulter Life Sciences, Eppendorf AG, Sigma-Aldrich, and Sartorius AG extend technological backing to the pharmaceutical sector. These facilitators are instrumental in endorsing and ascertaining regulatory adherence for the apparatus employed within the process. They grasp the precise requisites of the pharmaceutical domain, proposing an assortment of cell centrifugation apparatus, from benchtop to large-scale production centrifuges. Additionally, they furnish maintenance, repair, and training services, as well as tailored solutions for distinct pharmaceutical production prerequisites such as high-throughput processing or specialized sample preparation.

Innovative centrifugation strategies have been mooted for the high-throughput DSP of recombinant proteins, such as “centrifugation capture” (Roush & Lu, 2008), which endeavors to augment the efficiency and selectivity of protein purification utilizing centrifugal forces. The centrifugation capture technique involves functionalizing a solid support or affinity matrix with ligands specific to the target protein, which is then subjected to centrifugation to capture and separate the protein of interest from the bulk solution, thereby reducing the number of purification steps, diminishing costs, and enhancing the overall process efficiency.

Recently, methods entailing centrifugation have been illustrated for protein purification from *Escherichia coli*, such as utilizing HPH followed by a centrifugation step to eliminate cell debris (Li et al., 2013). Moreover, a distinct technique employing sucrose gradient ultracentrifugation has been described to segregate the inner and outer membranes of *E. coli*, which proves advantageous for subsequent membrane protein purification (Shu & Mi, 2023).

An integrated centrifugation process, as described in a study, explored the amalgamation of centrifugal partitioning chromatography and precipitation for monoclonal antibody purification, showcasing the potential of centrifugation in replacing chromatographic steps in protein purification processes (Oelmeier et al., 2013).

Furthermore, centrifugation can be amalgamated with other separation techniques such as tangential flow filtration (TFF) or ultrafiltration (UF) for more efficient recovery and concentration of target proteins. TFF proves advantageous for processing expansive volumes of culture broth, while UF aids in retaining and concentrating target proteins during the purification process (Ghosh, 2002).

Cumulatively, these evolutions in centrifugation techniques hold substantial promise for the DSP of recombinant proteins. By rendering the purification process more straightforward and cost-effective, these methods pose attractive substitutes for conventional purification techniques. With continuous research delving into novel centrifugation strategies, these techniques might gain increased relevance for large-scale protein production within the biotechnology and pharmaceutical sectors.

3.2 Filtration or tangential filtration

Tangential filtration is a versatile method used for both the recovery and purification of proteins in the biopharmaceutical industry (Luitjens & van Corven, 2019; Zydney & Kuriyel, 2000). It mainly differs in pore size, which is chosen based on the size and properties of the target protein, the complexity of the feed solution, and the desired purity and yield of the final product (Agrawal et al., 2023; Ordóñez, 2023). This method is employed in the recovery of intracellular proteins and even inclusion bodies, which are protein aggregates that can form during protein overexpression in host cells (Singh et al., 2015; Vallejo & Rinas, 2004).

Several major companies, such as Pall Corporation (2023), Merck KGaA (2023b), Sartorius AG (Sartorius, 2023), GE Healthcare Life Sciences (2013), 3M (2023), and MilliporeSigma (Merck KGaA, 2023a), provide filtration methods for the pharmaceutical industry. These companies offer a wide array of filtration products, such as TFF systems, depth filters,

TABLE 11.1 Companies providing filtration methods.

Company name	Filtration products offered
Pall Corporation	TFF systems, depth filters, sterile filtration products
Merck KGaA	TFF systems, depth filters, virus filters
Sartorius AG	TFF systems, depth filters, virus filters
GE Healthcare Life Sciences	TFF systems, depth filters, chromatography products
3M	TFF systems, depth filters, membrane filters
MilliporeSigma	TFF systems, depth filters, virus filters

TABLE 11.2 Materials used in pharmaceutical filters.

Material	Characteristics	Common applications
Polypropylene	Chemical resistance, durability	Harsh chemicals, high pressures
Nylon	High strength, abrasion resistance	Particulate matter, other contaminants removal
Polysulfone	High chemical resistance, high-temperature tolerance	High purity, low extractables applications
Cellulose/acetate	High flow rates, low protein binding	Protein recovery
PTFE	Excellent chemical resistance, low extractables	High purity, low protein binding applications

sterile filtration products, virus filters, and chromatography products (Table 11.1). The choice of filtration method will depend on the specific needs and requirements of the biopharmaceutical process (Harrison, 2014).

Filters used in the pharmaceutical industry are made from various materials, depending on the specific application and the type of contaminants that need to be removed (Pillai et al., 2016; Table 11.2). Some of the main materials used in filters for the pharmaceutical industry are polypropylene, nylon, polysulfone, cellulose acetate, and PTFE (Cytiva, 2023; Thermo Fisher Scientific, 2023a). Each of these materials possesses unique properties that make them suitable for specific applications.

For instance, polypropylene is a thermoplastic material known for its chemical resistance and durability, making it an ideal material for filters that need to withstand harsh chemicals and high pressures (Jenke, 2017). Nylon, on the other hand, is a synthetic polymer commonly used in filters for its high strength and resistance to abrasion, making it suitable for filters that need to remove particulate matter and other contaminants (Hawach Scientific, 2023).

Polysulfone is a thermoplastic material often used in filters for the pharmaceutical industry due to its high chemical resistance and ability to withstand high temperatures (Tisch Scientific, 2023a). It is frequently used in applications where high purity and low extractables are required. Cellulose acetate, a natural polymer, is known for its high flow rates and low protein-binding properties, making it ideal for applications where protein recovery is important (Shokri & Adibkia, 2013). Finally, PTFE (polytetrafluoroethylene) is a synthetic fluoropolymer often used in filters for the pharmaceutical industry due to its excellent chemical resistance and low extractables, making it suitable for applications where high purity and low protein binding are required (Tisch Scientific, 2023b).

When using filtration methods in the pharmaceutical industry, there are several operational parameters that need to be considered to ensure optimal filtration performance (Table 11.3). Some of the main parameters are filter pore size, filter area, flow rate, pressure, temperature, and pH. Careful control of these operational parameters is crucial in optimizing the filtration process and achieving the desired level of purity and yield for the target production (Raman & Klima, 2019).

Filter pore size, for example, is a critical parameter in determining filtration performance (Viswanathan et al., 2021). Choosing a filter with a pore size that is too small can result in blockage of the filter and reduced flow rates while choosing a filter with a pore size that is too large can result in insufficient removal of contaminants (Viswanathan et al., 2021). The filter area is another important parameter that affects filtration performance. Increasing the filter area can increase the filtration capacity and flow rates, but it also increases the cost of the filtration process (EMW filtertechnik GmbH, 2023).

TABLE 11.3 Key operational parameters for optimal filtration performance.

Parameter	Impact
Filter pore size	Affects blockage and flow rates, determines contaminant removal efficiency
Filter area	Influences filtration capacity, flow rates, and cost
Flow rate	Impacts filtration efficiency and risk of blockage
Pressure	Affects filtration rate, capacity, and risk of filter failure
Temperature	Impacts rate of fouling and risk of bacterial growth
pH	Affects the stability of filter material and filtration performance

Other parameters, such as flow rate, pressure, temperature, and pH, can also significantly impact filtration performance. High flow rates can result in reduced filtration efficiency and an increased risk of filter blockage, while low flow rates can result in reduced process efficiency and increased processing time. Similarly, extreme pH values can affect the stability of the filter material and result in reduced filtration performance (Raman & Klima, 2019).

3.3 Sedimentation

Sedimentation serves as an effective method for the separation of particulate matter from a liquid medium based on gravitational forces or centrifugal acceleration (Wiesmann & Binder, 1982). The process is pivotal in the recovery and purification steps within the biopharmaceutical industry, particularly in the context of AMP production (Wiesmann & Binder, 1982). The technique's efficiency is contingent on several factors, including particle size, density, and the properties of the liquid medium (Gonzalez Rodriguez et al., 2023). By modifying the sedimentation conditions or employing flocculating agents, the process's effectiveness can be significantly enhanced (Fan et al., 2020; Melnikova et al., 2021).

Major companies such as Alfa Laval, Beckman Coulter, and Thermo Fisher Scientific provide sedimentation equipment and technology support to ensure optimal operation within regulatory compliance. Their offerings range from simple gravity sedimentation systems to more complex centrifugal sedimentation equipment, catering to a myriad of industrial needs.

Recent advancements have sought to augment sedimentation techniques, adapting them to the specific requirements of AMP recovery and purification (Naimah et al., 2018; Zhang et al., 2021). These innovations aim not only to improve purity and yield but also to reduce operational costs, which is crucial for large-scale production. Operational parameters such as settling time, temperature, pH, and the use of flocculants are essential considerations for optimizing sedimentation processes (Jover-Smet et al., 2017). The careful manipulation of these parameters is instrumental in achieving the desired separation efficiency, which in turn significantly impacts the purity and yield of the AMPs obtained. Furthermore, sedimentation can be integrated with other DSP techniques such as centrifugation and filtration to further refine the recovery and purification processes, thereby ensuring the production of high-quality AMPs ready for subsequent applications in the pharmaceutical sector (Agarwal et al., 2020).

3.4 Influencing factors on recovery efficiency

The efficacy of AMP recovery during DSP can be significantly influenced by various operational, material, and methodological factors.

- *Operational conditions:* The process milieu, encompassing pH, temperature, ionic strength, and buffer composition, plays a crucial role in the recovery efficiency during the purification phase (Yap et al., 2022).
- *Matrix characteristics:* The intrinsic properties of the purification matrix, including the level of contaminants or competing proteins, may alter the recovery efficiency (Yap et al., 2022).
- *Peptide attributes:* The distinct attributes of AMPs, such as charge, size, hydrophobicity, and stability, are pivotal factors that can influence recovery efficiency during purification (Suda et al., 2017).

- *Purification methodology*: The choice of purification technique, be it affinity chromatography, ion exchange chromatography (IEC), or gel filtration chromatography, can exhibit varying efficiency levels for different AMPs (Pereira et al., 2021).
- *Production scale*: The efficiency of recovery may vary with the scale of production since the economic aspects of large-scale purification diverge from those of small-scale purification (Martin et al., 2020).
- *Quality assurance measures*: Ensuring a minimal presence of impurities or contaminants through stringent quality control measures is imperative to enhance recovery efficiency (D'Hondt et al., 2014).
- *Equipment quality*: The efficiency of recovery is also contingent on the type and maintenance condition of the equipment utilized; well-maintained or modern equipment is likely to outperform outdated or poorly maintained counterparts (Z. Liu & Smith, 2021).
- *Protease activity*: The presence of proteases could potentially degrade AMPs, thereby affecting the recovery efficiency adversely. Employing protease inhibitors or operating under conditions unfavorable to protease activity could mitigate such detrimental effects (Zhang et al., 2021).

Each of these factors necessitates careful consideration and optimization to ensure an efficient recovery process, thereby facilitating the production of high-quality AMPs for pharmaceutical applications.

4 Concentration and purification of antimicrobial peptides

4.1 Precipitation methods

Concentrating and purifying AMPs is a nuanced process often leveraging precipitation, the second most prevalent method for protein purification (Li et al., 2012). Precipitation methods are vital for concentrating and initially purifying proteins amidst diverse buffers or extracts. They operate on the principle of altering the target protein's solubility through specific reagents that modulate the repulsive electrostatic forces between proteins.

Understanding the solubility attributes of the target peptide is crucial before selecting a precipitation method, which can be achieved by examining the peptide's amino acid composition (Huan et al., 2020). The solubility in aqueous solutions is dictated by the number and types of ionic charges borne by the peptide. Typically, peptides with a higher count of charged residues exhibit better solubility in aqueous solutions. Particularly, peptides tend to bear more charges within a pH range of 6–8 compared to pH 2–6; hence, they dissolve more efficiently in near-neutral pH conditions (Zhang et al., 2021).

Nonetheless, there are exceptions to these general guidelines. For instance, peptides with a high hydrophobic sequence propensity tend to form aggregates. The drive toward aggregation is primarily due to the sequence's hydrophobicity, although extensive hydrogen bonding networks also contribute to this tendency (Cherry et al., 2014; Pirtskhalava et al., 2013; Zapadka et al., 2017). The categorization of a peptide as basic, acidic, or neutral follows specific guidelines:

- Acidic residues (D, E, and C-terminal COOH) are assigned a value of -1 .
- Basic residues (K, R, and the N-terminal NH₂) are assigned a value of $+1$.
- H residue is assigned a value of $+1$ at pH <6 and zero at pH >6 .

Subsequently, the total number of charges at pH 7 for D, E, K, R, and C-terminal COOH, and C-terminal NH₂ is calculated to ascertain the overall net charge of the peptide (Lear & Cobb, 2016; Schindler et al., 2021).

The solubility of the target peptide can be tested based on the previously mentioned guidelines through the following strategies:

- If the peptide's overall net charge is negative, implying an acidic nature, and/or if the total number of charges at pH 7 is over 25% of the total residue count, a small quantity of 0.1 M ammonium bicarbonate should be added to dissolve the peptide. The solution should then be diluted with water to the desired concentration, ensuring the resulting pH is around 7, with adjustments made as necessary (GenScript, 2023; Merck KGaA, 2023c).
- If the peptide's overall net charge is positive, indicating a basic nature, and the total charge count at pH 7 is between 10% and 25% of the total residue count, a small quantity of 25% acetic acid should be added to dissolve the peptide. This should then be diluted with water to the desired concentration (McCarthy et al., 2023; Merck KGaA, 2023c).
- If the peptide's overall net charge is zero, denoting a neutral nature, and the total charge count is over 25% of the total residue count, the first strategy described should be followed. If the total charge count is between 10% and 25% of the total residue count, the use of organic solvents is recommended (Do et al., 2021; PeptideSynthetics, 2023).
- If the total charge count of the peptide is below 10% of the total residue count, employing organic solvents is advisable (Do et al., 2021; PeptideSynthetics, 2023).

The solvent selection should align with the assay's tolerance for that specific solvent. If the peptide remains undissolved, a stronger solvent might be required. Upon dissolving, especially in an organic solvent, the peptide should be diluted by slowly adding the solution to a buffered solution while maintaining gentle agitation (Beltran et al., 2018; Zhang et al., 2021).

For peptides that are hydrophobic, uncharged, or possess a high proportion of hydrophobic residues, organic solvents such as acetonitrile (ACN), dimethylsulfoxide (DMSO), or dimethylformamide (DMF) are suitable (Abboud et al., 2021; Mueller et al., 2020). The inclusion of chaotropic compounds, such as guanidine hydrochloride or urea, could disrupt hydrophobic interactions or mitigate peptide “gelling” by disbanding the hydrogen bonding network (Mokgehele et al., 2021; Timson, 2020).

Acid precipitation emerges as another effective technique. Each protein has an isoelectric point that varies with the medium's pH, causing structural alterations in the protein and leading to precipitation. Acid addition to solutions can destabilize the peptides' structure, inducing precipitation (McCarthy et al., 2023).

A distinct set of precipitation methods involves the utilization of salt or alcohol. Salt can enhance protein solubility at lower concentrations, known as salting-in, but induce aggregation and precipitation at higher concentrations, termed salting-out. Alcohol, conversely, diminishes protein hydration, instigating aggregation and precipitation by water removal and protein encirclement. Common solvents used include ethanol, methanol, and acetone (Baghalabadi et al., 2021; Du et al., 2015).

In every method, operational conditions can influence the protein's functionality. For instance, trichloroacetic acid (TCA) precipitation denatures proteins, making this method more suited for concentrating samples for downstream applications such as SDS-PAGE rather than preserving protein function (Sivaraman et al., 1997; Takenaka et al., 1971).

Contrarily, salt-induced precipitation can selectively purify specific proteins by manipulating salt concentrations, such as ammonium sulfate, to precipitate undesired proteins while retaining the desired ones in solution. This method doesn't denature proteins; thus, they can be stored in high-salt solutions for extended durations without concerns over bacterial contamination or protease activity (Haas et al., 2022; Kim & Bae, 2003; Lim et al., 2020; Remans et al., 2022).

Furthermore, in peptide precipitation, salts often collaborate with organic solvents. For most peptides, escalating the organic solvent percentage beyond 80% can enhance sample recovery through precipitation, with acetone or ACN being commonly employed organic solvents (Baghalabadi et al., 2021).

In alcohol-based precipitation, experiments should be conducted at lower temperatures to preserve the proteins' biological functions. Moreover, monitoring the pH and protein concentration is essential (Lim et al., 2020; Schilcher et al., 2013).

In every situation, the specifics of the peptide and the experiment need meticulous consideration. For instance, if the peptide remains undissolved, sonication might be necessary to fragment the solid peptide into smaller particles, enhancing solubilization (Pchelintsev et al., 2016; Ruan et al., 2013). If the peptide is merely suspended in the solution, a stronger solvent or a different approach may be warranted.

The endeavor of concentrating and purifying AMPs is a complex undertaking necessitating a thorough understanding of the peptide's properties along with judicious selection and application of solvents and techniques. By adhering to these guidelines, the effective and efficient isolation of these crucial compounds for further study and application can be achieved (Huan et al., 2020; Zhang et al., 2021).

4.2 Dialysis and desalting techniques

Traditional dialysis operates on the principle of selective molecular diffusion through permeable membranes, effectively segregating small molecules from their larger counterparts (Cai et al., 2015). Primarily, it's deployed to modify the salt (small molecule) constitution of a solution harboring macromolecules. For sample volumes ranging from 0.2 to 200 mL, dialysis is achieved by encasing the sample in standard dialysis tubing submerged in a buffer with the desired ionic composition and strength while ensuring continuous buffer agitation (Thermo Fisher Scientific, 2023b). However, for larger volumes, employing multiple dialysis bags is an option, although desalting or buffer exchange at an expansive scale is expedited through tangential flow diafiltration (a form of dialysis utilizing UF while maintaining constant volume) (Cai et al., 2015; Rajendran et al., 2021). In utilizing dialysis or UF methods, it is pivotal to ascertain the peptide size and choose the tubing cut-off at least three times smaller than the peptide size to preclude peptide loss (Thermo Fisher Scientific, 2023b).

Desalting, the process of salt extraction from a sample, and buffer exchange, the substitution of one buffer salt set with another, are seamlessly executed through size exclusion chromatography (SEC), alternatively referred to as gel filtration chromatography (Naimah et al., 2018; Striegel, 2022) (elaborated in the following section).

It is imperative to align the methods and materials employed with the peptide's characteristics and the overarching goals of the purification process to ensure optimal recovery and purity of the AMPs (Gan et al., 2021; Nayab et al., 2022).

4.3 Chromatography methods for purification

The discipline of high-performance liquid chromatography (HPLC) has evolved remarkably over the past quarter century, showcasing vast versatility in the isolation and purification of peptides from a diverse spectrum of sources, quantities, and complexities. The demarcation between peptides and proteins, hinging on the count of amino acid residues they encompass, may appear arbitrary. However, peptides are traditionally defined as molecules comprising 50 or fewer amino acid residues. As the molecular chain lengthens, the potential for defined secondary or tertiary structures burgeons, making the conformation an increasingly pivotal aspect, especially when peptides exceed a length of 20–35 residues. This conformational propensity often hinges on the internalization of nonpolar residues fostered by peptide chain folding. Furthermore, the presence of disulfide bridges can significantly modulate peptide conformation, thereby impacting retention behavior during HPLC procedures. Consequently, conscientious consideration of conformation is indispensable when delineating the conditions for chromatographic operations (Sewald & Jakubke, 2009).

A rigorous analysis of amino acid properties is imperative while selecting the chromatographic method suitable for a particular peptide. The side chains of amino acids are typically categorized based on their polarity, hydrophobicity, and charge status, each playing a crucial role in peptide separation. The myriad variations in the size, shape, and properties of these side chains invariably influence the separation efficiency of peptides, thereby necessitating a thorough understanding of these attributes (Hughes, 2013).

In HPLC, three major modalities have traditionally dominated peptide separations, namely, size-exclusion HPLC (SEC), ion-exchange HPLC (IEX), and reverse-phase HPLC (RP-HPLC). Each modality exploits distinct peptide characteristics—size, net charge, or hydrophobicity, respectively—to facilitate separation. Within these broad frameworks, the conditions of the mobile phase can be meticulously tailored to amplify the separation potential of a specific HPLC column, thereby optimizing the purification process (Mant & Hodges, 1996).

The materials employed in these chromatographic techniques also play a pivotal role in the efficiency and effectiveness of peptide purification. Silica-based packings have historically been the go-to materials, courtesy of their rigidity, which enables high linear flow velocities of mobile phases (Qiu et al., 2011). However, they come with a limitation of pH tolerance, ranging between 2.0 and 8.0, as silica supports tend to dissolve rapidly in basic eluents (Mant et al., 2007). On the flip side, column packings based on organic polymers are witnessing a surge in usage across all modes of HPLC due to their broad pH tolerance, often spanning from pH 0 to 14 (Mant et al., 2007). Among these, materials formed from cross-linked polystyrene divinyl benzene are most commonly employed (Huck & Bonn, 2005). Supports based on polymethacrylate or vinyl alcohol copolymers are also showing promise (Pasch et al., 2005).

SEC stands out as a potent method for separating proteins based on their size, proving invaluable in the initial separation of peptides and proteins within a mixture (Brusotti et al., 2018). This technique has shown its mettle, especially within the molecular weight range of 100–7000 Da, thereby forming an integral part of the peptide purification toolkit (Mant et al., 2007).

4.4 Peptide purification by ion exchange chromatography

IEX hinges on the electrostatic attraction between the molecular charge of biomolecules and the opposite charge of functional groups affixed to a stationary phase. This methodology is pivotal for the segregation of cationic and anionic microbial peptides based on their electric charge at a specific pH level. The binding affinity between molecules and the stationary phase is inversely proportional to the ionic strength of the buffer and can be fine-tuned by adjusting the pH and ionic strength (Ngere et al., 2023; Wallace & Rochfort, 2023).

Two primary categories exist within the IEC: anion exchange and cation exchange. Anion exchange chromatography, with its stationary phase laden with positive charges such as tertiary or quaternary amines, is adept at binding anionic molecules. Conversely, cation exchange chromatography employs a negatively charged stationary phase, typically featuring carboxymethyl or sulfopropyl groups, to attract cationic molecules (Cho et al., 2019).

The IEC methodology is especially pertinent for acidic and basic peptides, often employing salt gradients near neutral pH (Liu et al., 2022). The choice between cation and anion exchangers hinges on the stability of the target peptide or protein. For instance, a cation exchanger is chosen when the target protein is more stable below its isoelectric point (pI), while an anion exchanger is used when the stability is higher above the pI (Chakraborty et al., 2022; Khan, 2012).

Furthermore, the strength of the exchanger is influenced by the pH range within which the target protein remains stable. Strong exchangers are chosen for extremely acidic conditions, extending down to pH 2.0. The resolution of IEC is a function of the particle size and porosity of the matrix particles, where smaller pore sizes yield higher resolution, albeit at lower flow rates (Choupani Chaydarreh et al., 2022).

Buffers are instrumental in IEC, ensuring a stable pH environment and thereby facilitating optimal interactions between the molecules and the stationary phase (Mohan, 2006; Welch, 2023). The buffer ions should mirror the charge of the ion exchanger's functional groups, and the pH should be modulated between the pKa of the surface functional groups and the pI of the protein molecules. The ionic strength of the buffer significantly impacts protein adsorption; thus, the selection of an appropriate buffer is imperative for the success of the ion exchange procedure (Mohan, 2006; Parkes et al., 2014; Pasche et al., 2005).

With the advent of new materials such as monoliths and magnetic beads, IEC has seen enhanced versatility and efficiency, opening new panoramas for protein purification in diverse biological and biotechnological fields (Du et al., 2022; Gautam, 2022; Oberacker et al., 2019). This progression has also facilitated the purification of more complex protein mixtures (Aslam et al., 2017; Zhang et al., 2001; Zhou et al., 2020), showcasing the evolving potential of IEC in contemporary peptide and protein purification endeavors (Du et al., 2022).

4.5 Peptide purification by reversed-phase chromatography and hydrophobic interaction chromatography

Reversed-phase high-performance liquid chromatography (RP-HPLC) has established itself as an indispensable tool in the analysis and purification of biomolecules due to its ability to resolve polypeptides with nearly identical sequences, be they small peptides obtained through trypsin digestion or substantially larger proteins (Aguilar, 2004; Colin & Guiochon, 1977). Utilized for the separation of peptide fragments from enzymatic digests and the purification of both natural and synthetic peptides, preparative RP-HPLC shines in purifying synthetic peptides in milligram-to-gram quantities (Aguilar, 2004). The essence of RP-HPLC lies in the continuous partitioning of molecules between the mobile phase and the hydrophobic stationary phase, separating polypeptides based on subtle differences in their “hydrophobic foot.” These differences arise from variations in amino acid sequences and conformations (Aguilar, 2004; Colin & Guiochon, 1977).

In RP-HPLC, the core of protein separation hinges on the hydrophobic interaction between nonpolar solute molecules in the mobile phase and hydrophobic ligands in the stationary phase matrix (Kulkarni & Bose, 2022; Taghvaei & Smith, 2020). The meticulous protocol for RPC encompasses the selection of the buffer system and ion-pairing agents, equilibration of the column, binding of the extract to the column, elution of protein molecules, cleaning and storage of the column, and the finalization of the purification process (Tetzlaff et al., 2012). The purity of the eluted protein is ascertained using SDS-PAGE (Zhou et al., 2020).

Transitioning to hydrophobic interaction chromatography (HIC), this technique excels in the separation and purification of hydrophobic molecules, capitalizing on hydrophobic interactions between the matrix and analyte. Unlike RPC, HIC performs better in a polar and less denaturing environment. The process of HIC entails protein loading to bind to the adsorbent under optimal pH and salt concentrations, the removal of nonspecifically bound proteins through buffer washes, and the separation of the target protein from impurities via a buffer with a decreasing salt concentration gradient (Fleming, 2020; O'Connor & Cummins, 2017).

The HIC column's adsorbent matrix significantly influences the degree of hydrophobic interaction. The type and concentration of the salt, along with the presence of additives, can significantly impact the purification process (Kulkarni & Bose, 2022). With a well-defined protocol for column eluate collection and adsorbent washing, HIC stands as an effective method for isolating hydrophobic proteins, playing a pivotal role in the post-manufacturing of biopharmaceutical products.

The discourse on RPC and HIC elucidates the pivotal role these chromatographic techniques play in peptide purification. Each technique, with its own unique set of principles and protocols, contributes significantly to the DSP of AMPs, underscoring the versatility and efficacy of chromatographic methods in peptide purification endeavors (Kailasam, 2021; Pilely et al., 2022).

5 Conclusions

The DSP of AMPs encompasses a spectrum of intricately designed steps, each pivotal in ensuring the purity and functionality of the peptides. Through methodologies such as centrifugation, sedimentation, filtration, dialysis, ion exchange, and chromatographic techniques, a meticulous orchestration of processes is achieved. The elucidation of each method, its principles, and its applications underscores the nuanced approach required in the purification and analysis of peptides. As the

realm of AMPs continues to burgeon, the refinement and advent of novel DSP techniques are quintessential, setting a trajectory toward enhanced efficacy in the purification processes and ultimately contributing to the advancement of the biopharmaceutical domain.

References

- 3M. (2023). Biopharmaceutical clarification and purification. https://www.3m.com/3M/en_US/biopharmaceutical-purification-us/applications/chemical-pharmaceutical-manufacturing/.
- Abboud, S. A., Cisse, E. H., Doudeau, M., Bénédetti, H., & Aucagne, V. (2021). A straightforward methodology to overcome solubility challenges for N-terminal cysteinyl peptide segments used in native chemical ligation. *Chemical Science*, 12(9), 3194–3201. 20416539. Royal Society of Chemistry, France. Available from: <http://pubs.rsc.org/en/journals/journal/sc>. <https://doi.org/10.1039/d0sc06001a>.
- Adje, E. Y., Balti, R., Lecouturier, D., Kouach, M., Dhulster, P., Guillochon, D., & Nedjar-Arroume, N. (2013). Controlled enzymatic hydrolysis: A new strategy for the discovery of antimicrobial peptides. *Proteomics and antimicrobial*. *Proteins*, 5(3), 176–186. 18671314 <https://doi.org/10.1007/s12602-013-9138-y>.
- Agarwal, A., Jaiswal, N., Tripathi, A. D., & Paul, V. (2020). *Downstream processing; applications and recent updates* (pp. 29–55). Springer Science and Business Media LLC. https://doi.org/10.1007/978-981-15-7070-4_2.
- Agrawal, P., Wilkstein, K., Guinn, E., Mason, M., Serrano Martinez, C. I., & Saylae, J. (2023). A review of tangential flow filtration: Process development and applications in the pharmaceutical industry. *Organic Process Research and Development*, 27(4), 571–591. 1520586X. American Chemical Society, United States. Available from: <http://pubs.acs.org/journal/oprdfk>. <https://doi.org/10.1021/acs.oprd.2c00291>.
- Aguilar, M. I. (2004). Reversed-phase high-performance liquid chromatography. *Methods in Molecular Biology (Clifton, N.J.)*, 251, 9–22. 10643745.
- Aslam, B., Basit, M., Nisar, M. A., Khurshid, M., & Rasool, M. H. (2017). Proteomics: Technologies and their applications. *Journal of Chromatographic Science*, 55(2), 182–196. 0021-9665 <https://doi.org/10.1093/chromsci/bmw167>.
- Baghalabadi, V., Razmi, H., & Doucette, A. (2021). Salt-mediated organic solvent precipitation for enhanced recovery of peptides generated by pepsin digestion. *Proteomes*, 9(4), 44. 2227-7382 <https://doi.org/10.3390/proteomes9040044>.
- Beltran, J. A., Aguilera-Mendoza, L., & Brizuela, C. A. (2018). Optimal selection of molecular descriptors for antimicrobial peptides classification: An evolutionary feature weighting approach. *BMC Genomics*, 19. 14712164. BioMed Central Ltd., Mexico. Available from: <http://www.biomedcentral.com/bmcgenomics>. <https://doi.org/10.1186/s12864-018-5030-1>.
- Brusotti, G., Calleri, E., Colombo, R., Massolini, G., Rinaldi, F., & Temporini, C. (2018). Advances on size exclusion chromatography and applications on the analysis of protein biopharmaceuticals and protein aggregates: A mini review. *Chromatographia*, 81(1), 3–23. 16121112. Friedr. Vieweg und Sohn Verlags GmbH, Italy. Available from: <http://www.chromatographia.de/>. <https://doi.org/10.1007/s10337-017-3380-5>.
- Cai, L., Yang, Y., Jiao, N., Zhang, R., & Almeida, A. (2015). Evaluation of tangential flow filtration for the concentration and separation of bacteria and viruses in contrasting marine environments. *PLoS ONE*, 10(8), e0136741. 1932-6203 <https://doi.org/10.1371/journal.pone.0136741>.
- Chakraborty, A., Puja, R., & Bose, K. (2022). *Protein purification by ion exchange chromatography textbook on cloning, expression and purification of recombinant proteins* (pp. 173–198). India: Springer Nature. Available from: <https://link.springer.com/book/10.1007/978-981-16-4987-5>. https://doi.org/10.1007/978-981-16-4987-5_7.
- Cherry, M. A., Higgins, S. K., Melroy, H., Lee, H.-S., & Pokorny, A. (2014). Peptides with the same composition, hydrophobicity, and hydrophobic moment bind to phospholipid bilayers with different affinities. *The Journal of Physical Chemistry B*, 118(43), 12462–12470. 1520-6106 <https://doi.org/10.1021/jp507289w>.
- Cho, G., Park, Y., Hong, Y. K., & Ha, D. H. (2019). Ion exchange: An advanced synthetic method for complex nanoparticles. *Nano Convergence*, 6(1). 21965404. Korea Nano Technology Research Society, South Korea Available from: <https://link.springer.com/journal/40580>. <https://doi.org/10.1186/s40580-019-0187-0>.
- Choupani Chaydarreh, K., Lin, X., Guan, L., & Hu, C. (2022). Interaction between particle size and mixing ratio on porosity and properties of tea oil camellia (*Camellia oleifera* Abel.) shells-based particleboard. *Journal of Wood Science*, 68(1). 16114663. Springer, China. Available from: <http://www.springerlink.com/content/1435-0211>. <https://doi.org/10.1186/s10086-022-02052-3>.
- Colin, H., & Guiochon, G. (1977). Introduction to reversed-phase high-performance liquid chromatography. *Journal of Chromatography A*, 141(3), 289–312. 00219673 [https://doi.org/10.1016/S0021-9673\(00\)93537-4](https://doi.org/10.1016/S0021-9673(00)93537-4).
- Cytiva. (2023). Membrane filtration: A guide to choosing membranes. <https://www.cytivalifesciences.com/en/us/news-center/quick-reference-membrane-compatibility-10001>.
- Desai, M. A. (2008). *Downstream processing of proteins: Methods and protocols. Downstream processing of proteins: Methods and protocols. Vol. 9.* Springer Science & Business Media.
- Dhiman, A., Thakur, K., Parmar, V., Sharma, S., Sharma, R., Kaur, G., Singh, B., & Suhag, R. (2023). New insights into tailoring physicochemical and techno-functional properties of plant proteins using conventional and emerging technologies. *Journal of Food Measurement and Characterization*, 17(4), 3845–3873. 21934134. Springer, India. Available from: <https://www.springer.com/journal/11694>. <https://doi.org/10.1007/s11694-023-01919-3>.
- D'Hondt, M., Bracke, N., Taevernier, L., Gevaert, B., Verbeke, F., Wynendaele, E., & De Spiegeleer, B. (2014). Related impurities in peptide medicines. *Journal of Pharmaceutical and Biomedical Analysis*, 101, 2–30. 1873264X. Elsevier, Belgium. Available from: <http://www.elsevier.com/locate/jpba>. <https://doi.org/10.1016/j.jpba.2014.06.012>.
- Ding, Y., & Kan, J. (2017). Optimization and characterization of high pressure homogenization produced chemically modified starch nanoparticles. *Journal of Food Science and Technology*, 54(13), 4501–4509. 09758402, Springer India, China. Available from: <http://www.springerlink.com/content/121580/>. <https://doi.org/10.1007/s13197-017-2934-8>.

- Do, H. T., Franke, P., Volpert, S., Klinksiek, M., Thome, M., & Held, C. (2021). Measurement and modelling solubility of amino acids and peptides in aqueous 2-propanol solutions. *Physical Chemistry Chemical Physics*, 23(18), 10852–10863. 14639076. Royal Society of Chemistry, Germany. Available from: <http://pubs.rsc.org/en/journals/journal/cp>. <https://doi.org/10.1039/d1cp00005e>.
- Du, M., Hou, Z., Liu, L., Xuan, Y., Chen, X., Fan, L., Li, Z., & Xu, B. (2022). Progress, applications, challenges and prospects of protein purification technology. *Frontiers in Bioengineering and Biotechnology*, 10, 22964185. Frontiers Media S.A., China. Available from: <http://journal.frontiersin.org/journal/bioengineering-and-biotechnology#archive>. <https://doi.org/10.3389/fbioe.2022.1028691>.
- Du, Y., Wu, D., Wu, Q., & Guan, Y. (2015). Quantitative evaluation of peptide-extraction methods by HPLC–triple-quad MS–MS. *Analytical and Bioanalytical Chemistry*, 407(6), 1595–1605. 1618–2642 <https://doi.org/10.1007/s00216-014-8389-0>.
- EMW filtertechnik GmbH. (2023). What role does effective filter media area play in air filtration?. <https://www.emw.de/en/filter-campus/effective-filter-media-area.html>.
- Fan, Y., Ma, X., Dong, X., Feng, Z., & Dong, Y. (2020). Characterisation of floc size, effective density and sedimentation under various flocculation mechanisms. *Water Science and Technology*, 82(7), 1261–1271. 0273–1223 <https://doi.org/10.2166/wst.2020.385>.
- Ferdous, S., Dopp, J. L., & Reuel, N. F. (2021). Optimization of *E. coli* tip-sonication for high-yield cell-free extract using finite element modeling. *AIChE Journal*, 67(10). <https://doi.org/10.1002/aic.17389>. 0001–1541.
- Fleming, R. (2020). ADC Analysis by Hydrophobic Interaction Chromatography. *Methods in Molecular Biology*, 2078, 147–161. Humana Press Inc., United States. Available from: <http://www.springer.com/series/7651>. https://doi.org/10.1007/978-1-4939-9929-3_10.
- Fox, J. L. (2013). Antimicrobial peptides stage a comeback. *Nature Biotechnology*, 31(5), 379–382. <https://doi.org/10.1038/nbt.2572>.
- Gan, B. H., Gaynord, J., Rowe, S. M., Deingruber, T., & Spring, D. R. (2021). The multifaceted nature of antimicrobial peptides: Current synthetic chemistry approaches and future directions. *Chemical Society Reviews*, 50(13), 7820–7880. 14604744. Royal Society of Chemistry, United Kingdom. Available from: <http://pubs.rsc.org/en/journals/journal/cs>. <https://doi.org/10.1039/d0cs00729c>.
- Gautam, A. (2022). *Magnetic bead-based nucleic acid isolation* (pp. 111–117). Springer Science and Business Media LLC. https://doi.org/10.1007/978-3-030-94230-4_15.
- GE Healthcare Life Sciences. (2013). Quality matters Whatman™ filtration products for pharmaceutical quality control. https://www.fishersci.de/content/dam/fishersci/en_EU/promotions/GEHealthcareLifeSciences/Whatman_Filtration_Quality_Matters_Pharma.pdf.
- GenScript. (2023). Guidelines for dissolving peptides. https://www.genscript.com/site2/document/12464_20100407015038.PDF.
- Ghosh, R. (2002). Protein separation using membrane chromatography: Opportunities and challenges. *Journal of Chromatography A*, 952(1–2), 13–27. 00219673 [https://doi.org/10.1016/S0021-9673\(02\)00057-2](https://doi.org/10.1016/S0021-9673(02)00057-2).
- Giovannoni, M., Gramegna, G., Benedetti, M., & Mattei, B. (2020). Industrial use of cell wall degrading enzymes: The fine line between production strategy and economic feasibility. *Frontiers in Bioengineering and Biotechnology*, 8, 22964185. Frontiers Media S.A., Italy. Available from: <http://journal.frontiersin.org/journal/bioengineering-and-biotechnology#archive>. <https://doi.org/10.3389/fbioe.2020.00356>.
- Goldberg, S. (2021). Mechanical/physical methods of cell disruption and tissue homogenization. *Methods in Molecular Biology*, 2261, 563–585. Humana Press Inc., United States. Available from: <http://www.springer.com/series/7651>. https://doi.org/10.1007/978-1-0716-1186-9_36.
- Gonzalez Rodriguez, L., McCallum, A., Kent, D., Rathnayaka, C., & Fairweather, H. (2023). A review of sedimentation rates in freshwater reservoirs: Recent changes and causative factors. *Aquatic Sciences*, 85(2), 14209055, Springer Science and Business Media Deutschland GmbH, Australia. Available from: <https://www.springer.com/journal/27>. <https://doi.org/10.1007/s00027-023-00960-0>.
- Haas, S., Desombre, M., Kirschhöfer, F., Huber, M. C., Schiller, S. M., & Hubbuch, J. (2022). Purification of a hydrophobic elastin-like protein toward scale-suitable production of biomaterials. *Frontiers in Bioengineering and Biotechnology*, 10, 22964185. Frontiers Media S.A., Germany. Available from: <http://journal.frontiersin.org/journal/bioengineering-and-biotechnology#archive>. <https://doi.org/10.3389/fbioe.2022.878838>.
- Harrison, S. T. L. (2011). *Comprehensive biotechnology* 2.44 - cell disruption. Elsevier.
- Harrison, R. G. (2014). Bioseparation basics. *Chemical Engineering Progress*, 110(10), 36–42. Available from: <https://www.aiche.org/sites/default/files/cep/20141036a.pdf>.
- Hawach Scientific. (2023). Features and applications of nylon filter membrane. <https://www.hawachmembrane.com/features-and-applications-of-nylon-filter-membrane/>.
- Hielscher Ultrasonics GmbH. (2023a). Industrial ultrasonic devices. <https://www.hielscher.com/industry.htm>.
- Hielscher Ultrasonics GmbH. (2023b). Scale-up of ultrasonic extraction processes. <https://www.hielscher.com/scale-up-of-ultrasonic-extraction-processes.htm>.
- Huan, Y., Kong, Q., Mou, H., & Yi, H. (2020). Antimicrobial peptides: Classification, design, application and research progress in multiple fields. *Frontiers in Microbiology*, 11, 1664302X. Frontiers Media S.A., China. Available from: <https://www.frontiersin.org/journals/microbiology>. <https://doi.org/10.3389/fmicb.2020.582779>.
- Huck, C. W., & Bonn, G. K. (2005). Poly(styrene-divinylbenzene) based media for liquid chromatography. *Chemical Engineering and Technology*, 28(12), 1457–1472. 09307516 <https://doi.org/10.1002/ceat.200500265>.
- Hughes, A. B. (2013). *Amino acids, peptides and proteins in organic chemistry, analysis and function of amino acids and peptides*. John Wiley & Sons.
- Jagschies, G., Lindskog, E., Lacki, K., & Galliher, P. M. (2018). *Biopharmaceutical processing: Development, design, and implementation of manufacturing processes*. Elsevier.
- Jasmi, N., Mansor, N., Lim, E. J., Yusof, N. L., Hajar-Azhari, S., & Rahim, M. H. A. (2020). The effect of sonication and heat treatment on the physicochemical, nutritional and microbiological properties of different sugarcane variants. *Food Science and Technology*, 40(3), 551–556. 1678–457X <https://doi.org/10.1590/fst.12619>.
- Jenke, D. (2017). Extractables screening of polypropylene resins used in pharmaceutical packaging for safety hazards. *PDA Journal of Pharmaceutical Science and Technology*, 71(5), 346–367. <https://doi.org/10.5731/pdajpst.2016.007401>.

- Jover-Smet, M., Martín-Pascual, J., & Trapote, A. (2017). Model of suspended solids removal in the primary sedimentation tanks for the treatment of urban wastewater. *Water*, 9(6), 448. 2073–4441 <https://doi.org/10.3390/w9060448>.
- S. Kailasam. (2021). Technology networks liquid chromatography in the biopharmaceutical industry. <https://www.technologynetworks.com/analysis/articles/liquid-chromatography-in-the-biopharmaceutical-industry-354851>.
- Khan, H. U. (2012). *The role of ion exchange chromatography in purification and characterization of molecules* (pp. 331–342). InTech.
- Kim, S. G., & Bae, Y. C. (2003). Salt-induced protein precipitation in aqueous solution: Single and binary protein systems. *Macromolecular Research*, 11(1), 53–61. 15985032, Polymer Society of Korea, South Korea, Available from: <http://www.springerlink.com/content/121720/>. <https://doi.org/10.1007/BF03218278>.
- Kruszewski, B., Domian, E., & Nowacka, M. (2023). Influence of high-pressure homogenization on the physicochemical properties and Betalain pigments of red beetroot (*Beta vulgaris* L.) juice. *Molecules*, 28(5), 2018. 1420–3049 <https://doi.org/10.3390/molecules28052018>.
- Kula, M.-R., & Schütte, H. (1987). Purification of proteins and the disruption of microbial cells. *Biotechnology Progress*, 3(1), 31–42. 15206033 <https://doi.org/10.1002/btpr.5420030107>.
- Kulkarni, R., & Bose, K. (2022). *Protein purification by reversed phase chromatography and hydrophobic interaction chromatography. Textbook on cloning, expression and purification of recombinant proteins* (pp. 221–247). Springer Nature, India, Available from: <https://link.springer.com/book/10.1007/978-981-16-4987-5>. https://doi.org/10.1007/978-981-16-4987-5_9.
- Lear, S., & Cobb, S. L. (2016). Pep-Calc.com: A set of web utilities for the calculation of peptide and peptoid properties and automatic mass spectral peak assignment. *Journal of Computer-Aided Molecular Design*, 30(3), 271–277. 15734951. Springer International Publishing, United Kingdom. Available from: <https://www.springer.com/journal/10822>. <https://doi.org/10.1007/s10822-016-9902-7>.
- Levy, R., Okun, Z., & Shpigelman, A. (2021). High-pressure homogenization: Principles and applications beyond microbial inactivation. *Food Engineering Reviews*, 13(3), 490–508. 18667929. Springer, Israel. Available from: <http://www.springer.com/life+sci/food+science/journal/12393>. <https://doi.org/10.1007/s12393-020-09239-8>.
- Li, Y. (2011). Recombinant production of antimicrobial peptides in *Escherichia coli*: A review. *Protein Expression and Purification*, 80(2), 260–267. <https://doi.org/10.1016/j.pep.2011.08.001>.
- Li, G., Liu, B., Shang, Y., Yu, Z., & Zhang, R. (2012). Novel activity evaluation and subsequent partial purification of antimicrobial peptides produced by *Bacillus subtilis* LFB112. *Annals of Microbiology*, 62(2), 667–674. 18692044 <https://doi.org/10.1007/s13213-011-0303-9>.
- Li, Q., Mannall, G. J., Ali, S., & Hoare, M. (2013). An ultra scale-down approach to study the interaction of fermentation, homogenization, and centrifugation for antibody fragment recovery from *rec E. coli*. *Biotechnology and Bioengineering*, 110(8), 2150–2160. <https://doi.org/10.1002/bit.24891>.
- Lim, D. G., Lee, J. C., Kim, D. J., Kim, S. J., Yu, H. W., & Jeong, S. H. (2020). Effects of precipitation process on the biophysical properties of highly concentrated proteins. *Journal of Pharmaceutical Investigation*, 50(5), 493–503. 20936214. Available from: <http://www.springer.com/biomed/journal/40005>. <https://doi.org/10.1007/s40005-020-00471-2>.
- Liu, Y. D., Cadang, L., Bol, K., Pan, X., Tschudi, K., Jazayri, M., Camperi, J., Michels, D., Stults, J., Harris, R. J., & Yang, F. (2022). Challenges and strategies for a thorough characterization of antibody acidic charge variants. *Bioengineering*, 9(11), 23065354. Available from: <http://www.mdpi.com/journal/bioengineering>. <https://doi.org/10.3390/bioengineering9110641>.
- Liu, Z., & Smith, S. R. (2021). Enzyme recovery from biological wastewater treatment. *Waste and Biomass Valorization*, 12(8), 4185–4211. 1877265X. Springer Science and Business Media B.V., United Kingdom. Available from: <http://www.springer.com/engineering/journal/12649>. <https://doi.org/10.1007/s12649-020-01251-7>.
- Luitjens, A., & van Corven, E. (2019). *Production and purification of recombinant proteins pharmaceutical biotechnology: Fundamentals and applications* (pp. 57–82). Netherlands: Springer International Publishing. Available from: <http://www.springer.com/in/book/9783030007096>. https://doi.org/10.1007/978-3-030-00710-2_4.
- Mant, C. T., Chen, Y., Yan, Z., Popa, T. V., Kovacs, J. M., Mills, J. B., Tripet, B. P., & Hodges, R. S. (2007). HPLC analysis and purification of peptides. *Methods in Molecular Biology (Clifton, N.J.)*, 386, 3–55. 10643745 https://doi.org/10.1007/978-1-59745-430-8_1.
- Mant, C. T., & Hodges, R. S. (1996). Analysis of peptides by high-performance liquid chromatography. *Methods in Enzymology*, 271, 3–51. 00766879. Academic Press Inc., Canada. Available from: <https://www.sciencedirect.com/bookseries/methods-in-enzymology>. [https://doi.org/10.1016/s0076-6879\(96\)71003-0](https://doi.org/10.1016/s0076-6879(96)71003-0).
- Martin, V., Egelund, P. H. G., Johansson, H., Thordal Le Quement, S., Wojcik, F., & Sejer Pedersen, D. (2020). Greening the synthesis of peptide therapeutics: An industrial perspective. *RSC Advances*, 10(69), 42457–42492. 20462069. Royal Society of Chemistry, Denmark. Available from: <http://pubs.rsc.org/en/journals/journal/ra>. <https://doi.org/10.1039/d0ra07204d>.
- Maurin, O., Verdié, P., Subra, G., Lamaty, F., Martinez, J., & Métro, T. X. (2017). Peptide synthesis: Ball-milling, in solution, or on solid support, what is the best strategy? *Beilstein Journal of Organic Chemistry*, 13, 2087–2093. 18605397, Beilstein-Institut Zur Förderung der Chemischen Wissenschaften, France. Available from: <http://www.beilstein-journals.org/bjoc/content/pdf/1860-5397-13-206.pdf>. <https://doi.org/10.3762/bjoc.13.206>.
- McCarthy, D., Han, Y., Carrick, K., Schmidt, D., Workman, W., Matejtschuk, P., Duru, C., & Atouf, F. (2023). Reference standards to support quality of synthetic peptide therapeutics. *Pharmaceutical Research*, 40(6), 1317–1328. 1573904X. Springer, United States. Available from: <https://www.springer.com/journal/11095>. <https://doi.org/10.1007/s11095-023-03493-1>.
- Melnikova, A. A., Komova, A. V., Vasilev, R. A., Namsaraev, Z. B., & Gorin, K. V. (2021). Flocculation-based methods of microalgae removal from the eutrophic pond enrichment culture. *Water Supply*, 21(8), 4254–4262. 1606–9749 <https://doi.org/10.2166/ws.2021.174>.
- Merck KGaA. (2023a). Filtración para bioprocesos. <https://www.sigmaaldrich.com/CO/es/products/filtration/filtration-for-bioprocessing>.
- Merck KGaA. (2023b). Filtration methods to sterilize liquids. https://www.emdmillipore.com/US/en/life-science-research/chromatography-sample-preparation/filtration-methods/general-filtration/sterilization/iYSb.qB._MAAAFBRq9lvzJX.nav?ReferrerURL=https%3A%2F%2Fwww.google.com%2F&bd=1.

- Merck KGaA. (2023c). Solubility guidelines for peptides. <https://www.sigmaaldrich.com/CO/es/technical-documents/technical-article/research-and-disease-areas/cell-and-developmental-biology-research/solubility-guidelines>.
- Middelberg, A. P. J. (2000). *2 Microbial cell disruption by high-pressure homogenization* (pp. 11–21). Springer Science and Business Media LLC. https://doi.org/10.1007/978-1-59259-027-8_2.
- Mohan, C. (2006). *Buffers. A guide for the preparation and use of buffers in biological systems* (pp. 20–21). EMD Bioscience.
- Mokgehele, T., Madala, N., Gitari, W., & Tavengwa, N. (2021). Deciphering the effects of kosmotrope and chaotrope salts during aqueous two phase extraction (ATPE) of polyphenolic compounds and glycoalkaloids from the leaves of a nutraceutical plant, *Solanum retroflexum*, with the aid of UPLC-QTOF-MS. *South Africa Applied Biological Chemistry*, 64(1), 24680842. Springer Science and Business Media BV. Available from: <http://link.springer.com/journal/volumesAndIssues/13765>. <https://doi.org/10.1186/s13765-021-00603-8>.
- Mueller, L. K., Baumruck, A. C., Zhdanova, H., & Tietze, A. A. (2020). Challenges and perspectives in chemical synthesis of highly hydrophobic peptides. *Frontiers in Bioengineering and Biotechnology*, 8, 22964185, Frontiers Media S.A., Germany. Available from: <http://journal.frontiersin.org/journal/bioengineering-and-biotechnology#archive>. <https://doi.org/10.3389/fbioe.2020.00162>.
- Naimah, A. K., Al-Manhel, A. J. A., & Al-Shawi, M. J. (2018). Isolation, purification and characterization of antimicrobial peptides produced from *Saccharomyces boulardii*. *International Journal of Peptide Research and Therapeutics*, 24(3), 455–461. 15733904, Springer Netherlands, Iraq. Available from: <http://www.springeronline.com/sgw/cda/frontpage/0,11855,4-40109-70-35677487-0,00.html>. <https://doi.org/10.1007/s10989-017-9632-2>.
- Nayab, S., Aslam, M. A., Rahman, S., Sindhu, Z. D., Sajid, S., Zafar, N., Razaq, M., Kanwar, R., & Amanullah. (2022). A review of antimicrobial peptides: Its function, mode of action and therapeutic potential. *International Journal of Peptide Research and Therapeutics*, 28(1), 15733904. Springer Science and Business Media B.V., Pakistan. Available from: <http://www.springeronline.com/sgw/cda/frontpage/0,11855,4-40109-70-35677487-0,00.html>. <https://doi.org/10.1007/s10989-021-10325-6>.
- Nemer, G., Louka, N., Vorobiev, E., Salameh, D., Nicaud, J. M., Maroun, R. G., & Koubaa, M. (2021). Mechanical cell disruption technologies for the extraction of dyes and pigments from microorganisms: A review. *Fermentation*, 7(1), 23115637, MDPI AG, France. Available from: <https://www.mdpi.com/2311-5637/7/1/36/pdf>. <https://doi.org/10.3390/fermentation7010036>.
- Ngere, J. B., Ebrahimi, K. H., Williams, R., Pires, E., Walsby-Tickle, J., & McCullagh, J. S. O. (2023). Ion-exchange chromatography coupled to mass spectrometry in Life science, environmental, and medical research. *Analytical Chemistry*, 95(1), 152–166. 15206882, American Chemical Society, United Kingdom. Available from: <http://pubs.acs.org/journal/ancham>. <https://doi.org/10.1021/acs.analchem.2c04298>.
- O'Connor, B., & Cummins, P. M. (2017). Hydrophobic interaction chromatography Brendan. *Methods in Molecular Biology*, 1485, 355–363. Humana Press Inc., Ireland. Available from: <http://www.springer.com/series/7651>. https://doi.org/10.1007/978-1-4939-6412-3_18.
- Oberacker, P., Stepper, P., Bond, D. M., Höhn, S., Focken, J., Meyer, V., Schelle, L., Sugrue, V. J., Jeunen, G.-J., Moser, T., Hore, S. R., von Meyenn, F., Hipp, K., Hore, T. A., & Jurkowski, T. P. (2019). Bio-on-magnetic-beads (BOMB): Open platform for high-throughput nucleic acid extraction and manipulation. *PLoS Biology*, 17(1), e3000107. 1545-7885 <https://doi.org/10.1371/journal.pbio.3000107>.
- Oelmeier, S. A., Ladd-Effio, C., & Hubbuch, J. (2013). Alternative separation steps for monoclonal antibody purification: Combination of centrifugal partitioning chromatography and precipitation. *Journal of Chromatography A*, 1319, 118–126. 18733778 <https://doi.org/10.1016/j.chroma.2013.10.043>.
- Ordóñez, R. (2023). A quick guide for membrane selection in tangential flow filtration - Bionet. Available [Bionet.com](https://www.bionet.com).
- Pall Corporation. (2023). *19 industrial manufacturing filtration solutions*. Pall Corporation. <https://www.pall.com/en/industrial-manufacturing.html?>.
- Parkes, M., Myant, C., Cann, P. M., & Wong, J. S. S. (2014). The effect of buffer solution choice on protein adsorption and lubrication. *Tribology International*, 72, 108–117. 0301679X <https://doi.org/10.1016/j.triboint.2013.12.005>.
- Pasch, H., Adler, M., Knecht, D., Rittig, F., & Lange, R. (2005). New chromatographic and hyphenated techniques for hydrophilic copolymers. In *Macromolecular symposia* (pp. 166–177). Wiley-VCH Verlag Germany. <https://doi.org/10.1002/masy.200590021>. 15213900 [http://onlinelibrary.wiley.com/journal/10.1002/\(ISSN\)1521-3900](http://onlinelibrary.wiley.com/journal/10.1002/(ISSN)1521-3900).
- Pasche, S., Vörös, J., Griesser, H. J., Spencer, N. D., & Textor, M. (2005). Effects of ionic strength and surface charge on protein adsorption at PEGylated surfaces. *Journal of Physical Chemistry B*, 109(37), 17545–17552. 15206106 <https://doi.org/10.1021/jp050431+>.
- Pchelintsev, N. A., Adams, P. D., & Nelson, D. M. (2016). Critical parameters for efficient sonication and improved chromatin immunoprecipitation of high molecular weight proteins. *PLoS ONE*, 11(1), e0148023. 1932-6203 <https://doi.org/10.1371/journal.pone.0148023>.
- PeptideSynthetics. (2023). General guide for dissolving peptides. <https://www.peptidesynthetics.co.uk/docs/SolventGuidelines.pdf>.
- Pereira, A. M., da Costa, A., Dias, S. C., Casal, M., & Machado, R. (2021). Production and purification of two bioactive antimicrobial peptides using a two-step approach involving an elastin-like fusion tag. *Pharmaceuticals*, 14(10), 14248247, MDPI, Portugal. Available from: <https://www.mdpi.com/1424-8247/14/10/956/pdf>. <https://doi.org/10.3390/ph14100956>.
- Pilely, K., Johansen, M. R., Lund, R. R., Kofoed, T., Jørgensen, T. K., Skriver, L., & Mørtz, E. (2022). Monitoring process-related impurities in biologics—host cell protein analysis. *Analytical and Bioanalytical Chemistry*, 414(2), 747–758. 16182650, Springer Science and Business Media Deutschland GmbH, Denmark. Available from: <https://link.springer.com/journal/216/volumes-and-issues>. <https://doi.org/10.1007/s00216-021-03648-2>.
- Pillai, S. A., Chobisa, D., Urimi, D., & Ravindra, N. (2016). Filters and filtration: A review of mechanisms that impact cost, product quality and patient safety. *Journal of Pharmaceutical Sciences and Research*, 8(5), 271. 0975-1459.
- Pirtskhalava, M., Vishnepolsky, B., & Grigolava, M. (2013). Transmembrane and antimicrobial peptides. Hydrophobicity, amphiphilicity and propensity to aggregation. *arXiv: Biomolecules*.
- Qiu, H., Liang, X., Sun, M., & Jiang, S. (2011). Development of silica-based stationary phases for high-performance liquid chromatography. *Analytical and Bioanalytical Chemistry*, 399(10), 3307–3322. 16182650 <https://doi.org/10.1007/s00216-010-4611-x>.

- Rahman, M. M., Hosano, N., & Hosano, H. (2022). Recovering microalgal bioresources: A review of cell disruption methods and extraction technologies. *Molecules*, 27(9), 2786. 1420-3049 <https://doi.org/10.3390/molecules27092786>.
- Rajendran, S. R. C. K., Mason, B., & Doucette, A. A. (2021). Review of membrane separation models and technologies: Processing complex food-based biomolecular fractions. *Food and Bioprocess Technology*, 14(3), 415–428. 19355149. Springer, Canada. Available from: <http://www.springer.com/life+sci/food+science/journal/11947>. <https://doi.org/10.1007/s11947-020-02559-x>.
- Ramadhan, K., & Foster, T. J. (2018). Effects of ball milling on the structural, thermal, and rheological properties of oat bran protein flour. *Journal of Food Engineering*, 229, 50–56. 02608774, Elsevier Ltd, United Kingdom. Available from: <http://www.sciencedirect.com/science/journal/02608774>. <https://doi.org/10.1016/j.jfoodeng.2017.10.024>.
- Raman, G. S. S., & Klima, M. S. (2019). Effects of temperature and pH on pressure filtration of coal refuse slurry. *Mining, Metallurgy & Exploration*, 36. <https://doi.org/10.1007/s42461-019-0088-0>.
- Remans, K., Lebendiker, M., Abreu, C., Maffei, M., Sellathurai, S., May, M. M., Vaněk, O., & de Marco, A. (2022). Protein purification strategies must consider downstream applications and individual biological characteristics. *Microbial Cell Factories*, 21(1). 14752859, BioMed Central Ltd, Germany. Available from: <http://www.microbialcellfactories.com/home/>. <https://doi.org/10.1186/s12934-022-01778-5>.
- Rios, G. M., Belleville, M.-P., & Paolucci-Jeanjean, D. (2007). Membrane engineering in biotechnology: Quo vamus? *Trends in Biotechnology*, 25(6), 242–246.
- Roca-Pinilla, R., Lisowski, L., Arís, A., & Garcia-Fruitós, E. (2022). The future of recombinant host defense peptides. *Microbial Cell Factories*, 21(1). 14752859, BioMed Central Ltd, Australia. Available from: <https://microbialcellfactories.biomedcentral.com/>. <https://doi.org/10.1186/s12934-022-01991-2>.
- Rodríguez, A., & Vanechoutte, M. (2019). Comparison of the efficiency of different cell lysis methods and different commercial methods for RNA extraction from *Candida albicans* stored in RNAlater. *BMC Microbiology*, 19(1). 14712180, BioMed Central Ltd., Belgium. Available from: <http://www.biomedcentral.com/bmcmicrobiol/>. <https://doi.org/10.1186/s12866-019-1473-z>.
- Roush, D. J., & Lu, Y. (2008). Advances in primary recovery: Centrifugation and membrane technology. *Biotechnology Progress*, 24(3), 488–495. <https://doi.org/10.1021/bp070414x>.
- Ruan, L., Luo, H., Zhang, H., & Xing, Z. (2013). Effect of sonication on a novel designed peptide. *Journal of Wuhan University of Technology Materials Science Edition*, 28(3), 622–626. 10002413 <https://doi.org/10.1007/s11595-013-0741-2>.
- Salazar, O., & Asenjo, J. A. (2007). Enzymatic lysis of microbial cells. *Biotechnology Letters*, 29(7), 985–994. 15736776 <https://doi.org/10.1007/s10529-007-9345-2>.
- Saranya, N., Devi, P., Nithiyanantham, S., & Jeyalaxmi, R. (2014). Cells disruption by ultrasonication. *BioNanoScience*, 4(4), 335–337. 21911649. Springer Science and Business Media, LLC, India. Available from: <https://www.springer.com/journal/12668>. <https://doi.org/10.1007/s12668-014-0149-2>.
- Sartorius. (2023). Sartorius filter devices tailored for the pharmaceutical industry. <https://www.sartorius.com/en/products/lab-filtration-purification/pharmaceutical-filtration>.
- Schilcher, G., Schlagenhauf, A., Schneditz, D., Scharnagl, H., Ribitsch, W., Krause, R., Rosenkranz, A. R., Stojakovic, T., Horina, J. H., & Beloin, C. (2013). Ethanol causes protein precipitation—New safety issues for catheter locking techniques. *PLoS ONE*, 8(12), e84869. 1932-6203 <https://doi.org/10.1371/journal.pone.0084869>.
- Schindler, O., Raček, T., Maršavelski, A., Koča, J., Berka, K., & Svobodová, R. (2021). Optimized SQE atomic charges for peptides accessible via a web application. *Czech Republic Journal of Chemistry*, 13(1). 17582946. BioMed Central Ltd. Available from: <https://jcheminf.biomedcentral.com/>. <https://doi.org/10.1186/s13321-021-00528-w>.
- Sewald, N., & Jakubke, H.-D. (2009). *Peptides: Chemistry and biology*. Wiley-VCH Verlag GmbH & Co. <https://doi.org/10.1002/9783527626038>.
- Shokri, J., & Adibkia, K. (2013). *Cellulose-medical, pharmaceutical and electronic applications. Application of cellulose and cellulose derivatives in pharmaceutical industries*. IntechOpen. <https://doi.org/10.5772/55178>.
- Shu, S., & Mi, W. (2023). Separating inner and outer membranes of *Escherichia coli* by EDTA-free sucrose gradient centrifugation. *Bio-Protocol*, 13(6). <https://doi.org/10.21769/BioProtoc.4638>.
- Singh, A., Upadhyay, V., Upadhyay, A. K., Singh, S. M., & Panda, A. K. (2015). Protein recovery from inclusion bodies of *Escherichia coli* using mild solubilization process. *Microbial Cell Factories*, 14(1). 14752859. BioMed Central Ltd., India. Available from: <http://www.microbialcellfactories.com/home/>. <https://doi.org/10.1186/s12934-015-0222-8>.
- Sivaraman, T., Kumar, T. K. S., Jayaraman, G., & Yu, C. (1997). The mechanism of 2,2,2-trichloroacetic acid-induced protein precipitation. *Journal of Protein Chemistry*, 16(4), 291–297. 02778033 <https://doi.org/10.1023/A:1026357009886>.
- Striegel, A. M. (2022). Size-exclusion chromatography: A twenty-first century perspective. *Chromatographia*, 85(4), 307–313. 16121112, Springer Science and Business Media Deutschland GmbH, United States. Available from: <http://www.chromatographia.de/>. <https://doi.org/10.1007/s10337-022-04143-1>.
- Suda, S., Field, D., & Barron, N. (2017). Antimicrobial peptide production and purification. *Methods in Molecular Biology*, 1485, 401–410. Humana Press Inc., Ireland. Available from: <http://www.springer.com/series/7651>. https://doi.org/10.1007/978-1-4939-6412-3_22.
- Sun, B., Wibowo, D., Middelberg, A. P. J., & Zhao, C. X. (2018). Cost-effective downstream processing of recombinantly produced pexiganan peptide and its antimicrobial activity. *AMB Express*, 8(1). 21910855. Springer Verlag, Australia. Available from: <http://www.amb-express.com/>. <https://doi.org/10.1186/s13568-018-0541-3>.
- Taghvaei, M., & Smith, B. (2020). Development and optimization of a reversed-phase HPLC method to separate pulse proteins. *Food Analytical Methods*, 13(7), 1482–1491. 1936976X, Springer, United States. Available from: <http://www.springer.com/life+sci/food+science/journal/12161>. <https://doi.org/10.1007/s12161-020-01771-x>.

- Takenaka, A., Takenaka, O., Mizota, T., Shibata, K., & Inada, Y. (1971). Sodium trichloroacetate as a denaturation reagent for proteins. *The Journal of Biochemistry*, 70(1), 63–73. 1756–2651 <https://doi.org/10.1093/oxfordjournals.jbchem.a129627>.
- Tetzlaff, J. M., Chan, A. W., Kitchen, J., Sampson, M., Tricco, A. C., & Moher, D. (2012). Guidelines for randomized clinical trial protocol content: A systematic review. *Systematic Reviews*, 1(1). 20464053. Available from: <http://www.systematicreviewsjournal.com/content/1/1/43>. <https://doi.org/10.1186/2046-4053-1-43>.
- Thermo Fisher Scientific. (2023a). Microporous filter membrane guide. <https://assets.thermofisher.com/TFS-Assets/LCD/Specification-Sheets/microporous-filter-membrane-guide-11-19-20.pdf>.
- Thermo Fisher Scientific. (2023b). Overview of dialysis, desalting, buffer exchange and protein concentration. <https://www.thermofisher.com/co/en/home/life-science/protein-biology/protein-biology-learning-center/protein-biology-resource-library/pierce-protein-methods/overview-buffer-exchange.html>.
- Timson, D. J. (2020). The roles and applications of chaotropes and kosmotropes in industrial fermentation processes. *World Journal of Microbiology and Biotechnology*, 36(6). 15730972, Springer, United Kingdom. Available from: <http://www.wkap.nl/journalhome.htm/0959-3993>. <https://doi.org/10.1007/s11274-020-02865-8>.
- Tisch Scientific. (2023a). Polysulfone membrane filters. <https://support.scientificfilters.com/product-applications/membrane-filter-selection/polysulfone-membrane-filters/>.
- Tisch Scientific. (2023b). PTFE filters: Characteristics, benefits, and common applications. <https://support.scientificfilters.com/filter-media/ptfe/>.
- Vallejo, L. F., & Rinas, U. (2004). Strategies for the recovery of active proteins through refolding of bacterial inclusion body proteins. *Microbial Cell Factories*, 3, 14752859. Available from: <http://www.microbialcellfactories.com/content/3/1/11>. <https://doi.org/10.1186/1475-2859-3-11>.
- Vinchhi, P., Patel, J. K., & Patel, M. M. (2021). High-pressure homogenization techniques for nanoparticles. *Emerging Technologies for Nanoparticle Manufacturing*, 263–285. Springer, India. Available from: <https://link.springer.com/book/10.1007/978-3-030-50703-9>. https://doi.org/10.1007/978-3-030-50703-9_11.
- Viswanathan, S., Stewart, M. L., & Rothamer, D. A. (2021). Experimental investigation of the effect of pore size distribution on nano-particle capture efficiency within ceramic particulate filters. *Emission Control Science and Technology*, 7(1), 26–40. 21993637. Springer Science and Business Media Deutschland GmbH, United States. Available from: <http://www.springer.com/earth+sciences+and+geography/engineering+geology/journal/40825/PSE>. <https://doi.org/10.1007/s40825-021-00184-4>.
- Wallace, R. G., & Rochfort, K. D. (2023). Ion-exchange chromatography: Basic principles and application. *Methods in Molecular Biology*, 2699, 161–177. Humana Press Inc., Ireland. Available from: <https://www.springer.com/series/7651>. https://doi.org/10.1007/978-1-0716-3362-5_9.
- Welch. (2023). Types of IEC buffers. <https://www.welch-us.com/types-of-iec-buffers/>.
- Wiesmann, U., & Binder, H. (1982). *Biomass separation from liquids by sedimentation and centrifugation* (pp. 119–174). Walter de Gruyter GmbH. <https://doi.org/10.1515/9783112620762-004>.
- Xiao, H., Liang, J., Zhang, Y., Chang, J., Zhang, R., & Zhang, P. (2023). Conversion of materials and energy in anaerobic digestion of sewage sludge with high-pressure homogenization pretreatment. *Processes*, 11(8), 2467. 2227–9717 <https://doi.org/10.3390/pr11082467>.
- Yadav, K. S., & Kale, K. (2020). High pressure homogenizer in pharmaceuticals: Understanding its critical processing parameters and applications. *Journal of Pharmaceutical Innovation*, 15(4), 690–701. 19398042, Springer, India. Available from: <https://link.springer.com/journal/12247>. <https://doi.org/10.1007/s12247-019-09413-4>.
- Yap, P. G., Lai, Z. W., & Tan, J. S. (2022). Bacteriocins from lactic acid bacteria: Purification strategies and applications in food and medical industries: A review. *Beni-Suef University Journal of Basic and Applied Sciences*, 11(1). 23148543, Springer Science and Business Media Deutschland GmbH, Malaysia. Available from: <http://bjbas.springeropen.com/>. <https://doi.org/10.1186/s43088-022-00227-x>.
- K.L. Zapadka, F.J. Becher, A.L. Gomes dos Santos, S.E. Jackson, Factors affecting the physical stability (aggregation) of peptide therapeutics. *Interface Focus* 7 (6) (2017), 20428901, Royal Society Publishing, United Kingdom, Available from: <http://rsfs.royalsocietypublishing.org/content/royfocus/7/6/20170030.full.pdf>. doi: <https://doi.org/10.1098/rsfs.2017.0030>.
- Zhang, H.-T., Kacharina, J. E., Miyashiro, K., Greene, M. I., & Eberwine, J. (2001). Protein quantification from complex protein mixtures using a proteomics methodology with single-cell resolution. *Proceedings of the National Academy of Sciences*, 98(10), 5497–5502. 0027-8424 <https://doi.org/10.1073/pnas.101124598>.
- Zhang, Q. Y., Yan, Z. B., Meng, Y. M., Hong, X. Y., Shao, G., Ma, J. J., Cheng, X. R., Liu, J., Kang, J., & Fu, C. Y. (2021). Antimicrobial peptides: Mechanism of action, activity and clinical potential. *Military Medical Research*, 8(1). 20549369, BioMed Central Ltd, China, Available from: <http://mmrjournal.biomedcentral.com/about>. <https://doi.org/10.1186/s40779-021-00343-2>.
- Zhou, X. F., Zhang, C. L., Gao, X. P., Wang, W. L., He, Z. F., Jiang, F. Y., Pang, Y. L., Li, J. H., Ren, X. J., Zhou, H. B., Tan, G. Q., Lyu, J. X., & Wang, W. (2020). A simple and rapid protein purification method based on cell-surface display of SUMO-fused recombinant protein and Ulp1 protease. *AMB Express*, 10(1). 21910855, Springer, China. Available from: <http://www.amb-express.com/>. <https://doi.org/10.1186/s13568-020-00999-4>.
- Zydney, A. L., & Kuriyel, R. (2000). *4 high-performance tangential flow filtration for protein separations* (pp. 35–46). Springer Science and Business Media LLC. https://doi.org/10.1007/978-1-59259-027-8_4.

ANTIMICROBIAL PEPTIDES

A Roadmap for Accelerating Discovery and Development

Edited by

Luis H. Reyes, Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

Juan C. Cruz, Department of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia

Gregory R. Wiedman, Department of Chemistry and Biochemistry Seton Hall University, South Orange, NJ, United States

Contains a comprehensive description of computational and experimental approaches for the discovery, manufacture, and clinical translation of antimicrobial peptides.

Antimicrobial Peptides: A Roadmap for Accelerating Discovery and Development covers the most important efforts of scientists and engineers worldwide to accelerate the process of discovery, production, and eventual market penetration of more potent antimicrobial peptides. These efforts have been fueled by emerging technologies such as artificial intelligence and data science, molecular and CFD simulations, easy-to-use process simulation packages, microfluidics, and 3D printing, among many others. Such technologies can now be implemented and scaled up quickly and at relatively low cost in low-budget production facilities, critical to moving to sustainable and marketable products worldwide.

Discovering novel antimicrobial peptides rationally and cost-effectively has emerged as one of the significant challenges of modern biotechnology. Thus far, this process has been tedious and costly, resulting in molecules with activities far below those needed to address the current challenge of microbial resistance to antibiotics that takes the lives of thousands of people around the world every year. Due to these issues, large biopharmaceutical companies have decided not to invest in discovering novel antimicrobials, which is worrisome in the long run due to the possibility of a new pandemic.

This book also delineates how multidisciplinary teams have assembled to address the challenges of manufacturing, biological testing, and clinical trials to finally reach complete translation. It is an ideal resource for scientists, researchers, engineers, and students in the fields of biotechnology, microbiology, biochemistry, and pharmaceuticals. Finally, anyone interested in the emerging technologies that are enabling the discovery and development of new antimicrobial peptides may find this book interesting and informative.

Key Features

- Covers computational tools (including emerging artificial intelligence algorithms) and microfluidic systems for discovering and high-throughput screening of AMPs
- Discusses the application of bioprocess engineering scale-up approaches for AMPs production and purification with the aid of process simulation tools and rapid prototyping
- Highlights user-centered design and formulation of products with AMPs
- Describes the whole pipeline for AMPs production



ELSEVIER

elsevier.com/books-and-journals

ISBN 978-0-443-15393-8



9 780443 153938